

Module 3 Project

**AI/ML-Based Predictive Maintenance and
Vehicle Health Monitoring System**

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1. Abstract

This project presents an AI/ML-based Predictive Maintenance and Vehicle Health Monitoring System designed to predict potential vehicle or component failure from operating parameters. The system uses engine temperature, RPM, oil pressure, vibration, battery voltage, coolant temperature, fuel consumption, vehicle speed, and operating hours as input parameters. A synthetic vehicle dataset was generated for model development and testing. Logistic Regression, Decision Tree, Random Forest, XGBoost, LSTM, GRU, and Transformer models were implemented for failure prediction. The final monitoring system estimates failure probability, converts it into a vehicle health score, assigns a vehicle status, generates a maintenance alert, and identifies major contributing factors.

2. Introduction

Predictive maintenance uses data-driven methods to identify abnormal operating conditions and potential failures before they occur. Vehicle operating parameters such as temperature, pressure, vibration, electrical voltage, speed, fuel consumption, and operating hours can provide useful information about vehicle condition.

The objective of this project is to develop a machine-learning and deep-learning based system that analyses vehicle operating parameters and predicts the possibility of failure. The prediction is converted into a vehicle health score and maintenance status.

3. Problem Statement

Unexpected vehicle or component failure can cause downtime, maintenance costs, and reduced reliability. This project addresses the problem by analysing vehicle operating parameters, predicting failure probability, calculating a vehicle health score, and generating a maintenance alert.

4. Objectives

- Develop a dataset containing important vehicle operating parameters.
- Preprocess vehicle sensor data for machine learning.
- Predict potential vehicle/component failure.
- Compare machine learning and deep learning models.
- Estimate failure probability.
- Calculate a vehicle health score.
- Generate Healthy, Warning, or Critical status.
- Generate a maintenance alert.
- Identify major contributing factors.

5. System Architecture

Vehicle Sensors → Data Preprocessing → ML/DL Model → Failure Probability → Vehicle Health Score → Vehicle Status → Maintenance Alert

The system receives vehicle operating parameters, preprocesses them, applies trained models, and converts the prediction into an interpretable health status and maintenance recommendation.

6. Dataset

The supplied dataset contains 10,000 records and 10 columns. It was synthetically generated for this project and contains the specified vehicle operating parameters together with the Failure target.

Parameter	Role
Engine Temperature	Vehicle operating input
RPM	Vehicle operating input
Oil Pressure	Vehicle operating input
Vibration	Vehicle operating input
Battery Voltage	Vehicle operating input
Coolant Temperature	Vehicle operating input
Fuel Consumption	Vehicle operating input
Vehicle Speed	Vehicle operating input
Operating Hours	Vehicle operating input
Failure	Target variable

Total missing values in the supplied dataset: 0.

Failure distribution: {0: 9609, 1: 391}

7. Input Parameters

- Engine Temperature
- RPM
- Oil Pressure
- Vibration
- Battery Voltage
- Coolant Temperature
- Fuel Consumption
- Vehicle Speed
- Operating Hours

8. Data Preprocessing

- The nine vehicle parameters were selected as input features.
- Failure was used as the target variable.
- The dataset was divided into training and testing data.
- Standardization was applied for the models requiring scaled input.
- Sequential data with sequence length 10 was prepared for LSTM, GRU, and Transformer models.

9. Machine Learning Models

Logistic Regression

Used as a baseline classification model for failure prediction.

Decision Tree

Used to classify failure conditions through decision-based rules.

Random Forest

An ensemble of decision trees used for failure prediction and feature-importance analysis.

XGBoost

A boosting-based classification model used for failure probability prediction and the final vehicle health monitoring function.

10. Deep Learning Models

LSTM

An LSTM network was trained on sequential vehicle operating data.

GRU

A GRU network was trained using sequential vehicle data.

Transformer

A Transformer-based model using multi-head self-attention was implemented for sequence-based failure prediction.

11. Model Evaluation

The models were evaluated using Accuracy, Precision, Recall, and F1 Score. The detailed numerical outputs and comparison plots are available in the submitted Colab notebook.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	See notebook output	See notebook output	See notebook output	See notebook output
Decision Tree	See notebook output	See notebook output	See notebook output	See notebook output
Random Forest	See notebook output	See notebook output	See notebook output	See notebook output
XGBoost	See notebook output	See notebook output	See notebook output	See notebook output
LSTM	See notebook output	See notebook output	See notebook output	See notebook output
GRU	See notebook output	See notebook output	See notebook output	See notebook output
Transformer	See notebook output	See notebook output	See notebook output	See notebook output

12. Vehicle Health Monitoring System

The final monitoring function accepts the nine vehicle operating parameters and uses the trained XGBoost model to estimate failure probability. The failure probability is expressed as a percentage and the vehicle health score is calculated as 100 minus the failure percentage.

The implemented status thresholds are: HEALTHY for a health score of 80% or above, WARNING for 60% to below 80%, and CRITICAL for below 60%.

13. Maintenance Alert

- HEALTHY: No immediate maintenance required.
- WARNING: Maintenance inspection recommended soon.
- CRITICAL: Immediate maintenance inspection required.

14. Major Contributing Factors

The project uses XGBoost feature importance to identify important vehicle parameters associated with the prediction. These are mapped to understandable descriptions such as high vibration, high engine temperature, abnormal oil pressure, low battery voltage, high coolant temperature, high fuel consumption, high RPM, high vehicle speed, and high operating hours.

15. Results and Discussion

The completed system demonstrates an end-to-end predictive maintenance workflow. Vehicle operating parameters are converted into machine-learning inputs, models are trained and evaluated, and the final monitoring function produces a failure probability, health score, status, maintenance alert, and contributing factors.

16. Conclusion

The project implements an AI/ML-based predictive maintenance and vehicle health monitoring workflow. A synthetic vehicle dataset was created using the specified vehicle parameters, followed by preprocessing and model development. Multiple machine learning and deep learning approaches were implemented. The final system converts predicted failure probability into a vehicle health score and maintenance status, providing an interpretable summary of vehicle condition.

17. Future Scope

- Integration with real vehicle sensors.
- Use of real-world vehicle maintenance datasets.
- Real-time vehicle health monitoring.
- IoT-based sensor integration.
- Deployment on an embedded or edge-computing platform.

18. Tools and Technologies

- Python
- Google Colab
- NumPy
- Pandas
- Scikit-learn
- XGBoost
- TensorFlow / Keras
- Matplotlib
- Seaborn

19. Project Files

- AI_ML_Vehicle_Predictive_Maintenance.ipynb – project implementation notebook.
- vehicle_predictive_maintenance_dataset.csv – dataset used by the project.

20. References

- Python documentation.
- NumPy documentation.
- Pandas documentation.
- Scikit-learn documentation.
- XGBoost documentation.
- TensorFlow / Keras documentation.

21. GitHub Link

<https://github.com/RohitChatterjee2/AI-ML-Based-Predictive-Maintenance-and-Vehicle-Health-Monitoring-System>